INCH-POUND

MIL-R-55099E(CR)
4 October 1990

Superseding
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MILITARY SPECIFICATION

RECEIVER, RADIO R-442()/VRC

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of frequency-modulated (fm) vehicular radio receiver which operates on any one of 920 channels within the frequency range of 30.00 to 75.95 Megahertz (MHz) designated as Receiver, Radio R-442()/VRC. (See 6.1 and 6.7).

1.2 Appendix. Appendix, "Production Burn-in Plan for Receiver, Radio R-442()VRC", forms a part of this specification.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TD,Fort Monmouth, New Jersey 07703-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

ASMC N/A FSC 5820

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SPECIFICATIONS

MILITARY

MIL-S-901	Shockproof Equipment, Class HI (High Impact) Shipload
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-I-17563	Impregnants for Aluminum, Copper, Iron, Magnesium and Zinc Alloy Castings

STANDARDS

MILITARY

MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-276	Impregantion of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 1911-5094.)

2.1.2 Other Government documents, drawings, and publications.
The following other Government documents, drawings, and publications form apart of this documents to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

COMMUNICATIONS-ELECTRONICS COMMAND

SC-CL-57991	Drawing and Gage list for Receiver, Radio R-442()/VRC
SC-CL-323124	Electrical Test Fixtures for Radio Set AN/VRC-12
SC-DL-414900	Drawing and Data List for Receiver, Radio R-442()/VRC

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish five sample units to be subjected to first article inspection (see 6.4 in accordance with 4.3).

3.2 Construction. Receiver, Radio R-442()/VRC shall be constructed in accordance with the requirements of this specification, and of Drawing and Data List SC-DL-414900.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein, shall apply.

3.4 Castings. Casting shall conform to requirements 21 of MIL-STD-454.

3.4.1 Quality of die castings (see 4.4). Quality of die castings shall be as follows:

a. Surface conditions shall be clean, dry and free from chips.

b. Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered and hidden by paint.

c. Flash must be removed by grinding or filing.

d. The casting after impregnation shall be air tight, and no leakage shall be allowed when tested in accordance with 4.7.4 and 4.10.

e. Flatness shall be as specified in Government-furnished drawings.

f. Dimensions shall be as specified in the Government-furnished drawings.

3.4.2 Impregnation of aluminum castings. All exterior aluminum castings shall be impregnated in accordance with Standard MIL-STD-276 using Type II impregnating material in accordance with Specification MIL-I-17563. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231 (see 4.4).

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

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3.5.3 Serial numbers. Receiver, Radio R-442()/VRC shall be serial numbered.

3.6 Air-seal. When the equipment is tested as specified in 4.10, the decrease in vacuum shall not exceed 0.01 pounds per square inch per minute.

3.7 Interchangeability and gaging. Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. Replaceable modules shall meet the gage requirements of 4.12. Mechanical gaging of inaccessible modules shall be performed before assembly.

3.8 Service conditions. The equipment shall meet the following service conditions.

3.8.1 Temperature.

3.8.1.1 Low. During testing as specified in 4.7.1.1, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.9.1.1	Muting	4.9.1.1
3.9.1.2	Loudspeaker Output	4.9.1.2
3.9.1.3	Headphone Output	4.9.1.3
3.9.1.4	Monitor Output	4.9.1.4
3.9.5	Sensitivity	4.9.5
3.9.6.1	Tone Squelch Operation	4.9.6.1
3.9.6.2	Noise Squelch Operation	4.9.6.2
3.9.7	Catching Range	4.9.7
3.9.2.1	Narrow Band Distortion	4.9.2.1

After testing, the equipment shall meet full specification performance for these measurements.

3.8.1.2 High. During testing as specified in 4.7.1.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.9.1.1	Muting	4.9.1.1
3.9.1.2	Loudspeaker Output	4.9.1.2
3.9.1.3	Headphone Output	4.9.1.3
3.9.1.4	Monitor Output	4.9.1.4
3.9.5	Sensitivity	4.9.5
3.9.6.1	Tone Squelch Operation	4.9.6.1
3.9.6.2	Noise Squelch Operation	4.9.6.2
3.9.7	Catching Range	4.9.7
3.9.2.1	Narrow Band Distortion	4.9.2.1

After testing, the equipment shall meet full specification performance for

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these measurements.

3.8.2 Humidity (moisture resistance). During testing as specified in 4.7.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.9.1.1	Muting	4.9.1.1
3.9.1.2	Loudspeaker Output	4.9.1.2
3.9.1.3	Headphone Output	4.9.1.3
3.9.1.4	Monitor Output	4.9.1.4
3.9.5	Sensitivity	4.9.5
3.9.6.1	Tone Squelch Operation	4.9.6.1
3.9.6.2	Noise Squelch Operation	4.9.6.2
3.9.7	Catching Range	4.9.7
3.9.2.1	Narrow Band Distortion	4.9.2.1

After testing, the equipment shall meet full specification performance for these measurements.

3.8.3 Altitude. The equipment shall meet the requirements of 3.12 when tested as specified in 4.7.3.

3.8.4 Leakage (immersion). The equipment shall show no evidence of leakage after it is tested as specified in 4.7.4.

3.8.5 Orientation. The equipment shall meet the requirements of 3.12 when tested as specified in 4.7.5.

3.8.6 Sand and dust. The equipment shall meet the requirements of 3.12 when tested as specified in 4.7.6.

3.8.7 Vibration. When vibrated and operated in accordance with 4.7.7, the peceiver, Radio R-442()/VRC shall be capable of receiving a voice modulated signal as displayed on a Tektronic type 504 Oscilloscope or equal without break-up, loss of signal or clicks. After vibration as specified in 4.7.7, the items shall be inspected for any structural failure, mechanical binding, or loose parts such as screws, clamps, bolts, and nuts.

3.8.8 Bounce and shock. The equipment shall be capable of meeting the requirements of Table I.

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TABLE I. Bounce and shock.

Requirement	Test Para.	Performance after test
Bounce	4.7.8	Specified performance (Note A) No physical damage, except for paint abrasion.
Shock; Ballistic	4.7.9	Specified performance (Note A) No physical damage, except for paint abrasion.

NOTE A: After completion of the tests, the equipment shall meet specified performance for the following: Audio power (3.9.1), Distortion (3.9.2), Sensitivity (3.9.5), Squelch (3.9.6), and Catching Range (3.9.7).

3.9 Electrical requirements.

3.9.1 Audio power output.

3.9.1.1 Muting (see 4.9.1.1). The muting level of the muting circuit for the associated loudspeaker shall be 50 milliwatts ± 20 percent.

3.9.1.2 Loudspeaker output (see 4.9.1.2). The receiver audio amplifier shall deliver a minimum of 500 milliwatts into a 600 ohm noninductive load. At 22 VDC, input voltage and exposure to 150° F ambient for a minimum of one hour, loudspeaker output shall not be below a minimum of 350 milliwatts.

3.9.1.3 Headphone output (see 4.9.1.3). The headphone output shall supply a minimum of 100 milliwatts into a 600 ohm noninductive load. At 22 VDC, input voltage and exposure to 150° F ambient for a minimum of one hour, headphone output shall not be below a minimum of 85 milliwatts.

3.9.1.4 Monitor output (see 4.9.1.4). The monitor output, into a 150 ohm noninductive load, shall be 0.16 volt minimum and 0.31 volt maximum.

3.9.1.5 Wide band output (see 4.9.1.5). The wide band circuit shall deliver 1 milliwat ± 1.0 db when receiving a signal having 8 kHz deviation, and 10 dbm ± 1.0 db when receiving a signal having 25 kHz deviation.

3.9.2 Distortion.

3.9.2.1 Narrow band distortion (see 4.9.2.1). Overall total narrow-band harmonic distortion shall not exceed 8 percent.

3.9.2.2 Wide band distortion (see 4.9.2.2). Overall total harmonic distortion shall not exceed 5 percent with the narrow band IF filter removed and a circuit jumper substituted.

3.9.3 Volume control (see 4.9.3). The volume control shall simultaneously vary the loudspeaker and headphones output levels. At the minimum setting of

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the volume control, there shall be not more than 0.25 milliwatt (mw) output at the loudspeaker terminal, and not more than 0.06 mw output at the headset terminal. At the maximum setting of the control, the loudspeaker and headset levels shall be not less than the levels specified in 3.9.1.2 and 3.9.1.3. Variation of the volume control settings shall maintain the monitor output level within the limits specified in 3.9.1.4, and the maximum variation in monitor output level shall be less than 2 db. The wide band output shall not be affected by the adjustment of the volume control.

3.9.4 Limiting (see 4.9.4). The limiting characteristic of the receiver when measured in accordance with 4.9.4 shall be such that the audio power output shall not vary more than ± 1 db.

3.9.5 Sensitivity (see 4.9.5). RF signal inputs of 0.5 microvolts at 25.5 volts and 30.0 volts equipment inputs at room ambient temperature shall produce a signal-plus-noise-distortion to noise-plus-distortion ratio of 10 db minimum. RF signal inputs of 0.5 microvolts at 22.0 volts equipment input at room ambient temperature shall produce a S + N + D/N + D ratio of 8 db minimum. RF signal inputs of 0.6 microvolts at -40° F and 150° F shall produce a S + N + D/N ration of 10 db minimum for 25.5 volts and 30 volts equipment inputs and a S + N + D/N + D ration of 8 db minimum for 22 volts equipment input. This requirement does not apply for frequencies of 33.90, 45.20, 56.50 and 67.80 MHz.

3.9.6 Squelch sensitivity.

3.9.6.1 Tone operation (see 4.9.6.1). The tone operated squelch shall open with an rf signal of 0.5 microvolt or less. At the temperature extremes (-40° F and 150° F), the tone squelch shall operate at 0.8 microvolt or less. Faulty operation of the squelch shall not occur when the incoming signal is voice modulated. The squelch once having opened, shall remain open when the rf signal is reduced to 0.4 microvolt. At the temperature extremes (-40° F and 150° F), the squelch once having opened, shall remain open when the rf signal is reduced to 0.7 microvolt. With no rf signal applied to the receiver, the maximum tine for the call light to go out shall be one second, the loudspeaker output shall be zero, and the retransmit contact on the squelch relay shall be open. When the rf signal is applied, the call lamp shall light, the loudspeaker output shall appear, and the retransmit contact on the squelch relay shall produce a ground.

3.9.6.2 Noise operation (see 4.9.6.2). The noise operated squelch shall open with an rf signal of 0.7 microvolt or less. At the temperature extremes (-40° F and 150° F), the noise squelch shall operate at 0.8 microvolt or less. Faulty operation of the squelch shall not occur when the incoming signal is voice modulated. The squelch, once having opened, shall remain open when the rf signal is reduced to 0.55 microvolt. At the temperature extremes (-40° F and 150° F), the squelch, once having opened, shall remain open when the rf signal is reduced to 0.7 microvolt. With no rf signal applied to the receiver, the maximum time for the call light to go out shall be 4 seconds, the loudspeaker output shall be zero, and the retransmit contact on the squelch relay shall be open. When the rf signal is applied, the call lamp shall light, the loudspeaker output shall appear, and the retransmit contact on the squelch relay shall produce a ground.

3.9.7 Catching range (see 4.9.7). The sum of the plus and minus catching

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ranges shall not be less than 850 kHz, nor shall either the plus or minus ranges be less than 250 kHz.

3.9.8 Maximum signal plus noise to noise ration(see 4.9.8). The maximum signal plus noise to noise ratio shall not be less than 45 db.

3.9.9 Desensitization (see 4.9.9). A 25 millivolts interfering signal which is ± 500 kHz removed from the desired signal shall not degrade a 26 db signal plus noise to noise signal by more than 6 db.

3.9.10 Residual phase detector output (see 4.9.10). The residual phase detector output voltage shall be 0.17 vdc or less.

3.9.11 Input voltage range (see 4.9.11). Unless otherwise specified herein, the receiver shall meet full specification requirements when operating from a 22.0 to 30.0 vdc source.

3.9.12 Audio response.

3.9.12.1 Narrow band response (see 4.9.12.1). The overall audio response of the receiver shall be within ± 2 db of 1000 Hz reference from 500 to 3000 Hz.

3.9.12.2 Wide band response (see 4.9.12.2). The overall audio response of the receiver shall be within + 2 db, - 3 db of the 1000 Hz reference from 300 to 20,000 Hz.

3.9.13 VFO modulation sensitivity (see 4.9.13). The deviation sensitivity of the oscillator shall be within the limits of 300 kHz/V minimum and 600 kHz/V maximum, when measured in accordance with 4.9.13.

3.9.14 VFO deviation capability (see 4.9.14). The deviation of the oscillator shall not be less than 450 kHz nor greater than 950 kHz over the tuning range of the oscillator, when measured in accordance with 4.9.14.

3.9.15 VFO stability vs temperature (see 4.9.15). The local oscillator shall not vary more than ± 180 kHz including the 85 kHz allowed for resetting. The variation shall be within ± 130 kHz if frequencies are not reset.

3.9.16 VFO resetability (see 4.9.16). When measured in accordance with 4.9.16, the dial resetability shall be such that the maximum frequency difference between any two of 5 settings to any point on the dial from either direction shall not be greater than 85 kHz. When operation of the megahertz shorting switch is tested in accordance with 4.9.16, the minimum difference of the oscillator frequencies shall be greater than 150 kHz.

3.9.17 Output circuit impedance.

3.9.17.1 Loudspeaker output (see 4.9.17.1). The impedance of this circuit shall be 160 ohms ± 20 percent.

3.9.17.2 Headphone output (see 4.9.17.2). The impedance of this circuit shall be 43 ohms ± 20 percent.

3.9.17.3 Monitor output (see 4.9.17.3). The minimum output impedance of this circuit shall be 1800 ohms.

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3.9.17.4 Wide band audio output (see 4.9.17.4). The output impedance of this circuit shall be 600 ohms ± 20 percent.

3.9.18 IF selectivity (see 4.9.18). The IF selectivity shall conform to the following:

3.9.18.1 Narrow band.

a. Bandwidth 6 db points 32 kHz minimum.

b. Bandwidth 60 db points 80 kHz maximum.

c. Bandwidth 85 db points, 100 kHz maximum.

3.9.18.2 Wide band. With the narrow band filter removed and a circuit jumper substituted, IF selectivity shall be as follows:

a. Bandwidth 6 db points, 650 kHz minimum.

b. Bandwidth 40 db points, 6000 kHz maximum.

The response characteristics in the pass band shall have no irregularities greater than 2.5 db.

3.9.19 Spurious responses (see 4.9.19). There shall be not more than 10 spurious responses beyond ± 500 kHz of the desired frequency which are less than 100 db down from a one microvolt reference, and no more than 10 spurious responses with a ± 500 kHz of the desired frequency which are less than 80 db down from a one microvolt reference.

3.9.20 IF rejection (see 4.9.20). IF rejection shall not be less than 100 db.

3.9.21 Intermodulation (see 4.9.21). Any intermodulation resulting from two undesired carrier frequencies of the same level shall be at least 58 db down from the desired response. At the operating temperature extremes (-40° F and 150° F), any intermodulation shall be at least 56 db down from the desired response.

3.9.22 Oscillator radiation (see 4.9.22). The radiation of any oscillator within the equipment shall produce an output no greater than 75 microvolts when the receiver is tuned to any frequency within its tuning range. This requirement does not apply for frequencies of 33.90, 45.20, 56.50 and 67.80 MHz.

3.10 Preconditioning.

3.10.1 Bounce preconditioning. Each equipment on order shall be bounce preconditioned prior to electrical preconditioning and shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.10.2 Electrical preconditioning. Each equipment on order shall be electrically preconditioned in accordance with the Appendix of this

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specification.

3.11 500-hour operational requirement (see 4.13). The receiver shall operate continuously for a period of 500 hours at room ambient temperature and at 30 vdc input voltage, with no replacement of parts or realinement other than adjustment of front panel controls. At each 100 hour interval and at the conclusion of the 500 hour operational test,the receiver shall meet the required tests specified in 4.13.

3.12 Operational inspection. Each equipment on order shall be tested as specified in 4.14 immediately prior to final pack and packaging.

3.13 Electromagnetic Interference (EMI). The equipment shall be tested for compliance with MIL-STD-461 (see 6.2h). The test shall be performed on one first article sample and shall have no bearing on the acceptance of the preproduction sample test report (see 4.15). The following test methods are applicable: CE04, CE06, CS01, CS02, CS06, RE02, RE02.1, and RS03.2.

3.14 Technical literature and running spare parts. Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.15 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268

Paragraph	Inspection
3.8	Adhesives
3.10.7	Wires
3.15	Potting, encapsulation and embedment
3.16	Fibrous material, organic
3.21	Plastic material and parts
3.27	Finish, protective
3.30	Tropicalization of material and parts
3.31	Wiring and cabling
3.39	Controls
3.50	Lamps, lampholders, and light
3.57	Printed wiring assemblies
3.61	Screws, other threaded devices, and related parts
3.62	Tightening of threaded parts
3.63	Lockwashers
3.64	Riveting
3.65	Brazing, soldering, and welding
3.67	Anti-seize compound
3.72	Terminals, fixed and terminal boards

b. In this specification:

3.4	Castings
3.5	Marking

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First Article Inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, the inspection specified for Group A, Group B, and Group C (see Tables III, IV and V respectively), and 4.16. The electromagnetic interference test of 4.16 shall be for information only. The environmental tests shall be performed in the order shown in Table II according to the number of samples specified in the contract or order.

TABLE II. Order of environmental tests.

Test	Number of Samples				
	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Non-damaging tests					
Temperature (4.7.1)	1	1			
Immersion (4.7.4)			1	1	
Altitude (4.7.3)	2	2			
Orientation (4.7.5)			2	2	

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TABLE II. Order of environmental tests - Continued.

Test	Number of Samples				
	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Dust (4.7.6)	3	3			
Potentially Damaging Tests					
Humidity (4.7.2)	4	4			
Vibration (4.7.7.1)			3	3	
Bounce (4.7.8)			4	4	
Shock, ballistic (4.7.9)			5	5	
Electromagnetic Interference (4.16)				1	

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Parts, Materials and Processes Used in Electronic Equipment	3.3
Castings	3.4
Marking	3.5
Workmanship	3.15

4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be subjected to Bounce and Electrical Preconditioning (see para. 3.10) prior to inspection. The contractor shall perform the inspections specified in para 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce or electrical preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. The test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit of each lot equipment shall be inspected for conformance to all the inspection and test requirements of Table III, Group A Inspection. Group A inspection shall be performed in any order which is satisfactory to the Government, except the Air-Seal Test (4.10) shall be next to last test and Operational Inspection (4.14) shall be last.

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TABLE III. Group A inspection.

Inspection	Req Para	Insp Para
Visual and mechanical	3.15	4.11
A1000 VHF Tuner & Parent Board		
A1100 Antenna Board		
A1200 1st RF Board		
A1300 2nd RF Board		
A1400 Mixer Board		
A1500 Local oscillator		
A4100 1st & 2nd IF amplifier		
A4200 3rd, 4th, & 5th IF limiter and discriminator		
A4300 Audio & Squelch preamplifier		
A3100 CRS harmonic generator		
A3200 CRS balanced mixer		
A3400 CRS 1st & 2nd IF amplifier		
A3500 CRS 3rd IF amplifier & limiter		
A3600 CRS hunt discriminator		
A3700 CRS phase discriminator		
A5100 Audio amplifier		
A5200 Squelch amplifier		
A5300 Squelch filter		
A2100 Power supply board		
A2000 Crystal switch assembly		
A100 RF tuner assembly		
A4000 Low frequency rf assembly		
A3000 CRS assembly		
A5000 Audio & squelch assembly		
Final front panel assembly		
Assembled receiver without covers		
Assembled receiver with covers		
Electrical		
Distortion, narrow band	3.9.2.1	4.9.2.1
Volume Control	3.9.3	4.9.3
Limiting	3.9.4	4.9.4
Input Voltage	3.9.11	4.9.11
Air Seal Test	3.6	4.10
Operational Inspection	3.12	4.14

4.5.2 Group B inspection. This inspection shall conform to Table IV. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

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TABLE IV. - Group B inspection.

Inspection	Req Para	Insp Para
Maximum signal plus noise to noise ratio	3.9.8	4.9.8
Desensitization	3.9.9	4.9.9
Residual phase detector output	3.9.10	4.9.10
Audio response, narrow band	3.9.12.1	4.9.12.1
VFO modulation sensitivity	3.9.13	4.9.13
VFO deviation capability	3.9.14	4.9.14
VFO stability vs temperature	3.9.15	4.9.15
VFO resetability	3.9.16	4.9.16
Inspection by Gages (dimensions)	3.7	4.12(a)
Inspection by Gages (module performance)	3.7	4.12(b)

4.5.2.1 Group B sampling plans. The group B sampling plans is as shown in Table V.

Table V. Group B sampling plan.

Electrical		Inspection by gages (mechanical)	
Lot size	Sample size	Dimensions Sample size	Module Performance Sample size
2 to 8	3	2	5
9 to 15	3	2	5
16 to 25	3	3	5
26 to 50	5	5	5
51 to 90	6	5	7
91 to 150	7	6	11
151 to 280	10	7	13
281 to 500	11	9	16
501 to 1200	15	11	19
1201 to 3200	18	13	23
3201 to 10000	22	15	29

Note: Acceptance number in all cases is zero.

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on sample units that have been subjected to and met Groups A and B inspection.

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TABLE VI. Group C Inspection.

Inspection	Req Para	Insp Para
Subgroup I		
Immersion	3.8.4	4.7.4
Wideband tests		
Audio power output, wide band output	3.9.1.5	4.9.1.5
Distortion, wide band	3.9.2.2	4.9.2.2
Audio response, wide band	3.9.12.2	4.9 12.2
Wide band output impedance	3.9.17.4	4.9.17.4
IF selectivity, wide band	3.9.18.2	4.9.18
Subgroup III		
Loudspeaker output impedance	3.9.17.1	4.9.17.1
Headphone output impedance	3.9.17.2	4.9.17.2
Monitor output impedance	3.9.17.3	4.9.17.3
Subgroup IV		
IF selectivity, narrow band	3.9.18.1	4.9.18
Spurious responses	3.9.19	4.9.19
IF rejection	3.9.20	4.9.20
Intermodulation	3.9.21	4.9.21
Oscillator radiation	3.9.22	4.9.22
Subgroup V		
500 hour operational test	3.11	4.13
Bounce	3.8.8	4.7.8
Temperature	3.8.1	4.7.1
Vibration	3.8.7	4.7.7
Subgroup VI		
Humidity (Moisture Resistance)	3.8.2	4.7.2
Subgroup VII		
Altitude	3.8.3	4.7.3
Orientation	3.8.5	4.7.5
Sand and dust	3.8.6	4.7.6
Shock, Ballistic	3.8.8	4.7.9

4.5.3.1 Sampling for inspection of equipment. Units selected for each group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroups I, III, and IV. For these subgroups, two units from the first production lot shall be selected for the first group C inspection. For subsequent group C inspection, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

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4.5.3.1.2 Subgroup V. For this subgroup, two units from the first production lot shall be selected for the first group C inspection. For subsequent group C inspections, two units per month shall be selected.

4.5.3.1.3 Subgroup VI. For this subgroup, two units from the first production lot shall be selected for the first group C inspection. For subsequent group C inspections, two units every two months shall be selected.

4.5.3.1.4 Subgroup VII. For this subgroup, two units from the first production lot shall be selected for the first group c inspection. For subsequent group C inspections, two units every six months shall be selected.

4.5.3.2 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. The receiver shall then be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute.

4.6.2 Electrical preconditioning. All receivers, Radio R-442 ( )/VRC on order shall be electrically preconditioned in accordance with the burn-in plan delineated in the Appendix of this specification.

4.7 Service conditions tests.

4.7.1 Temperature test.

4.7.1.1 Low Temperature. The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -80° F and the operating temperature shall be -40° F. Prior to testing, the equipment shall fully comply with the performance requirements specified in

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equipment shall fully comply with the performance requirements specified in 3.8.1.1.

4.7.1.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be +150° F. Prior to testing, the equipment shall fully comply with the performance requirements specified in 3.8.1.2.

4.7.2 Humidity (moisture resistance) test. The equipment shall be tested in accordance with Method 507, Procedure III, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.3 Altitude test. The receiver, mounted on Mounting MT-1898 ( )/VRC, shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810. The operational test of 4.14 shall be performed before, during and after the altitude test to determine compliance with 3.8.3.

4.7.4 Leakage (immersion) test. The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure 1, of MIL-STD-810, to determine compliance with 3.8.4.

4.7.5 Orientation test. The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 (+3, -1) degrees. During inclination in each plane, the equipment shall be tested to 4.14 to determine compliance with 3.8.5.

4.7.6 Sand and dust test. The receiver shall be tested in accordance with Method 510, Procedure I, of MIL-STD-810. The operational test of 4.14 shall be performed before, during and after dust testing to determine compliance with 3.8.6.

4.7.7 Vibration. To determine compliance to 3.8.7, the equipment shall be measured as follows:

a. Secure the receiver directly to a vibration table that can be controlled within 10 percent of the specified amplitude.

b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch. Vibrate the equipment by sweeping at a constant rate from 10 to 55 cycles per second and then form 55 to 10 cycles per second. The total sweep up and down shall take 7 1/2 minutes per plane.

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c. Performance. To determine receive capability during the vibration test, the equipment shall be operated over the vibration frequency range specified in 4.7.7.1 (b) for all three directions to determine compliance with 3.8.7.1.

4.7.8 Bounce vehicular. The receiver, mounted on Mounting MT-1898 ( )/ VRC shall be tested in accordance with Method 514, Procedure IX, Part 2 of MIL-STD-810. During the bounce test, the receiver shall be operated under the standard test conditions of paragraph 4.8. After completion of the bounce test, the equipment shall be tested to determine compliance with 3.8.8.

4.7.9 Shock test; ballistic. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The receiver mounted on Mounting MT-1898 ()/VRC shall be secured in its normal operating position to the steel test plate by means of the same fasterners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows one each 1-foot blow, 3-feet blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. After completion of the shock test, the equipment shall be tested to determine compliance with 3.8.8.

4.8 Standard test conditions. All electrical tests shall be performed as follows:

a. In a screen room conforming to Specification MIL-STD-461.

b. The electrical tests specified in 4.7.1, 4.7.2, 4.7.5, 4.7.7, 4.7.8, and 4.7.9 shall be performed in interference free area with necessary precautions taken to prevent ambient interference from entering or affecting either the measuring circuit or equipments under test.

c. Under the following standard test conditions, unless otherwise specified:

Temperature----------Room ambient
Humidity----------Room ambient
Input voltage----------25.5 vdc
RF input----------10 microvolts, 8 kHz deviation at 1 kHz modulation frequency.

Test frequency---------60.05 Megahertz
Squelch----------Off
Audio output level:
  Loudspeaker----------500 Milliwatts
  Headphone----------100 milliwatts
  Wideband----------0 dbm
  Audio lead impedance 600 ohms
Narrow band IF filter

4.9 Electrical tests.

4.9.1 Audio output power.

4.9.1.1 Muting. The receiver shall be operated under standard test

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conditions with the Volume Control set for 500 mw speaker output. When the muting circuit is energized, the loudspeaker lever shall decrease to 50 mw ±20 percent (see 3.9.1.1).

4.9.1.2 Loudspeaker output. With the Volume Control set fully clockwise, the power dissipated into a 600 ohm non-inductive load shall be measured to determine compliance with 3.9.1.2.

4.9.1.3 Headphone output. With the volume control set fully clockwise, the power dissipated into a 600 ohm non-inductive load shall be measured to determine compliance with 3.9.1.3.

4.9.1.4 Monitor amplifier output. With the volume control set for 500 mw speaker output, and the monitor amplifier terminated into a 150 ohm non-inductive load, the output voltage shall be measured to determine compliance with 3.9.1.4.

4.9.1.5 Wide band output. With the rf input at 200 uv, the receiver shall be operated under standard test conditions except the narrow band IF filter shall be removed and replaced with a dummy filter having the input and output tied together internally. The deviation shall be as specified herein. The power output shall be measured for deviations of 8 and 25 kc to determine compliance with 3.9.1.5.

4.9.2 Distortion.

4.9.2.1 Distortion, narrow band. The receiver shall be operated under standard test conditions. The distortion at the loudspeaker terminal shall be measured to determine compliance with 3.9.2.1.

4.9.2.2 Distortion, wide band. The receiver shall be operated under standard test conditions except the narrow-band IF filter shall be removed and replaced with a dummy filter having the input and output connected together internally. The RF input to the receiver shall be 200 uv. The distortion at the wide band audio output shall be measured at 1000, 5000, and 10,000 Hertz to determine compliance with 3.9.2.2.

4.9.3 Volume control. The receiver shall be operated under standard test conditions. The output of the loudspeaker, headphone, monitor, and wide band circuits shall be measured for full clockwise (cw) and counter clockwise (ccw) settings of the volume control to determine compliance with 3.9.3.

4.9.4 Limiting. the receiver shall be operated under standard test conditions except that the rf input shall be 1.0 microvolt. The signal generator output shall then be increased to 100,000 microvolts and the audio power output continuously observed. The test shall be conducted at frequencies of 30.00, 75.95, 65.55, 41.05, 53.00, and 52.10 megahertz, and in that order. The receiver shall comply with 3.9.4.

4.9.5 Sensitivity. The receiver shall be operated under standard test conditions except the rf input shall be 0.5 microvolt. The ratio of the signal plus noise plus distortion to noise plus distortion, expressed in db, shall be measured with an appropriate distortion analyzer. Measurements shall be made at frequencies of 30.00, 65.10, 41.05, 75.90, 52.95, 53.00, 41.50, and 64.50 Megahertz, in that order. At the temperature extremes (-40° F and 150°

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F), this test shall be performed with an rf input of 0.6 microvolts. Sensitivity shall comply with 3.9.5.

4.9.6 Squelch sensitivity.

4.9.6.1 Tone operation. The receiver shall be operated under standard test conditions except the rf input shall be 0.5 microvolt. The squelch shall be adjusted to the NEW SQUELCH ON position, and the signal generator modulated with an additional tone of 150 Hz, 3 kHz deviation. The rf signal shall be applied to the receiver and then removed from the receiver to determine compliance with 3.9.6.1. The test shall be conducted at frequencies of 30.00, 65.20, 41.30, 52.95, 53.00, and 75.95 megahertz, in that order.

4.9.6.2 Noise operation. The receiver shall be operated under standard test conditions except the rf input shall be 0.7 microvolt and the squelch control adjusted to the OLD SQUELCH ON position. The rf signal shall be applied to the receiver and then removed from the receiver to determine compliance with 3.9.6.2. The test shall be conducted at frequencies of 30.00, 65.40, 41.50, 53.60, 52.70, and 75.80 megahertz, in that order.

4.9.7 Catching range. The receiver shall be operated under standard test conditions. Disconnect all cables from the crystal switch assembly and connect to a crystal switch assembly in another receiver or to a test jig containing a crystal switch assembly. Tune the receiver under test to 30.90 MHz and the crystal switch to the appropriate corresponding frequency. Tune the dial of the receiver under test in 50 kHz increments higher in frequency. After each movement of the dial, momentarily disable (short) the control line and observe whether or not the test signal can be received. Record the last dial reading at which the receiver is capable of receiving the 30.90 MHz signal. Repeat this procedure by setting the MC dial to 31.90 MHz and moving the dial of the receiver under test in 50 kHz increments lower in frequency. This test shall be repeated at each of the following pairs of frequencies to determine compliance with 3.9.7:

(30.90, 31.90), (31.95, 32.95), (32.90, 33.90), (33.95, 34.95)
(34.90, 35.90), (35.95, 36.95), (36.90, 37.90), (37.95, 38.95)
(38.90, 39.90), (39.95, 40.95), (40.90, 41.90), (41.95, 42.95)
(42.90, 43.90), (43.95, 44.95), (44.90, 45.90), (45.95, 46.95)
(46.90, 47.90), (47.95, 48.95), (48.90, 49.90), (49.95, 50.95)
(50.90, 51.90), and (51.95, 52.95).

4.9.8 Maximum signal plus noise to noise ratio. The receiver shall be operated under standard test conditions except the rf input shall be 100 microvolts. The ratio, expressed in db, of the audio output with and without modulation shall be measured. The test shall be conducted at frequencies of 30.00, 65.70, 52.95, 75.85, 45.80, and 53.10 megahertz, in that order.

4.9.9 Desensitization. Apply an rf signal to the receiver at a level to produce a 26 db S + N/N ratio. An interfering signal which is ± 0.5 MHz removed from the desired rf signal shall be applied, and its level increased until the S + N/N ratio of the desired signal is reduced to 20 db. The level of the interfering signal shall be recorded to determine compliance with 3.9.9. The test shall be conducted at frequencies of 30.00, 52.95, 53.05, and 75.10 megahertz.

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4.9.10 Residual phase detection output. The control loop shall be open (error signal lead connected). Connect a dc VTVM so that the output of the phase detector only is measured. With the receiver dial set to 41.45 MHz and with a 250 millivolts rf signal applied into the Crystal Reference System high pass filter, measure the residual dc output voltage while varying the rf signal frequency from 52.55 to 53.35 MHz. Disregard the output reading at 52.95 MHz. The residual phase detector output shall comply with 3.9.10.

4.9.11 Input voltage tests. To determine conformance with 3.9.11, the tests listed below shall be conducted at input voltages of 22.0 and 30.0 vdc.

a. Audio output power (see 4.9.1.1, 4.9.1.2, 4.9.1.3, and 4.9.1.4).

b. Sensitivity (see 4.9.5).

c. Squelch sensitivity (see 4.9.6).

d. Catching range (see 4.9.7).

e. Distortion (see 4.9.2.1).

4.9.12 Audio response.

4.9.12.1 Audio response, narrow band. The receiver shall be operated under standard test conditions. The signal generator shall be modulated with an external audio oscillator and the deviation set to 8 kHz at 1000 Hz modulated frequency. The audio output of the receiver shall be measured for frequencies of 500, 1000, 2000, and 3000 Hz, to determine compliance with 3.9.12.1.

4.9.12.2 Audio response, wide band. The receiver shall be operated at standard test conditions except the wide band IF system shall be installed. The test shall be performed in the same manner as that of 4.9.12.1, except measurements shall be made at frequencies of 300, 1000, 3000, 5000, 10,000, and 20,000 Hz, to determine compliance with 3.9.12.2.

4.9.13 VFO modulation sensitivity. The deviation of the local oscillator from the nominal frequency shall be measured when ± 0.5 vdc is applied to the modulator input through a 250k ohm resistor. With the control line open (no voltage applied to the modulator input), measure the VFO free running frequency. These measurements shall be taken at nominal VFO frequencies of 42.00 MHz (53.50 dial frequency), 52.00 MHz (63.50 MHz dial frequency), and 63.00 MHz (74.50 MHz dial frequency) to determine compliance with 3.9.13.

4.9.14 VFO deviation capability. The deviation of the VFO from nominal frequency shall be measured when ± 1.5 vdc is applied to the modulator input through a 250K ohm resistor. With the control line open (no voltage applied to the modulator input), measure the VFO free running frequency. These measurements shall be taken at nominal VFO frequencies of 42.00 MHz (53.50 MHz dial frequency), 52.00 MHz (63.50 MHz dial frequency), and 63.00 MHz (74.50 MHz dial frequency) to determine compliance with 3.9.14.

4.9.15 VFO stability vs temperature (see 3.9.15). The automatic frequency control line of the VFO shall be grounded and the nominal frequency of the VFO measured at room temperature. The ambient temperature shall then be lowered to -40° C, and maintained at that temperature for one hour, and then the VFO

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frequency shall be measured. The ambient temperature shall then be raised to + 65° C, maintained for one hour, and the nominal VFO frequency shall be measured. At each temperature step, the nominal VFO frequency shall be measured at dial frequencies of 53.50, 63.50, and 74.50 MHz. This test shall be conducted at input voltages of 22.0, 25.5, and 30.0 volts dc. Temperature stability shall comply with 3.9.15.

4.9.16 VFO resetability. With the VFO control line grounded and no input, turn the kHz and MHz knobs at least 5 detents below the test frequency where possible. The kHz knob shall be turned up to .50 MHz followed by turning the MHz knob up to 31 MHz to give the test frequency of 31.50 MHz. The frequency of the VFO shall be measured. Repeat the above, except the receiver shall be tuned to 31.50 MHz from the direction of a higher frequency. This procedure shall be repeated five times from each direction. The test shall be repeated for dial frequencies of 41.50 and 51.50 MHz to determine compliance with 3.9.16. At each test frequency, the operations of the megacycle shorting switch shall be tested with a dc VTVM connected between TP 3002 and ground. The MHz knob shall be seated by rocking slightly, and the oscillator frequency recorded. The MHz knob shall be stressed in a CW direction until the shorting switch is actuated as indicated by a sudden increase to 16 vdc on the VTVM. At this knob position, the oscillator frequency shall be measured. The difference between the oscillator frequencies shall conform to 3.9.16. Repeat in a CCW direction at each test frequency.

4.9.17 Output circuit impedance. These tests shall be conducted under standard test conditions.

4.9.17.1 Loudspeaker output impedance. The loudspeaker output shall be terminated in 160 ohms and the audio voltage across the termination adjusted with the volume control to one (1) volt. When the load is removed, the open circuit voltage shall increase to 2 + 0.4 volts (see 3.9.17.1).

4.9.17.2 Headphone output impedance. The headphone output shall be terminated in 43 ohms and the audio voltage across the termination adjusted with the volume control to 0.5 volt. When the load is removed, the open circuit output voltage shall increase to 1 $\pm$ 0.2 volts (see 3.9.17.2).

4.9.17.3 Monitor output impedance. For this test, the 8 kHz deviation shall be adjusted for a convenient unclipped reference level voltage at the unterminated monitor output and the reference level voltage shall be measured. Terminate the monitor output with a load of 1800 ohms. The open circuit voltage shall be at least twice the loaded value (see 3.9.17.3).

4.9.17.4 Wide band output impedance. The wide band output shall be terminated in 600 ohms and the output voltage measured. With the load removed, the open circuit voltage shall be twice the loaded value $\pm$ 10 percent (see 3.9.17.4).

4.9.18 IF selectivity, narrow band or wide band. A signal shall be injected into the input of the IF and the rectified signal at the output of the fourth stage of the IF shall be monitored. Adjust the input until this signal is at least 6 db below saturation. With this signal level as a reference, adjust the input frequency and level to determine compliance with 3.9.18.1 and 3.9.18.2.

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4.9.19 Spurious response. The amount of quieting produced by an unmodulated rf carrier of 1.0 microvolt shall be used as a reference. The signal generator output shall be increased to 100,000 microvolts and the generator tuned throughout the range of 20 to 100 MHz. Those frequencies which produce the quieting equal to or greater than the reference shall be recorded to determine compliance with 3.9.19. The test shall be conducted at frequencies of 30.00 and 69.25 MHz.

4.9.20 IF rejection. The receiver shall be turned to 30.00 MHz. The amount of quieting produced by 1 microvolt shall be used as a reference. The signal generator shall then be turned to 11.5 MHz and its output raised until the amount of quieting is equal to the references. The number of microvolts shall be recorded to determine compliance with 3.9.20. The test shall be performed at frequencies of 30.00, 52.00, 54.00, and 75.00 MHz.

4.9.21 Intermodulation. The receiver shall be turned to 41.5 MHz and an unmodulated rf carrier of 1.0 microvolt shall be applied to the input of the receiver. The amount of quieting produced by this signal shall be used as a reference. Two generators, similar in all respects, tuned to 41.6 and 41.7 MHz, shall be coupled to the input of the receiver. The output of the generators shall be kept equal and simultaneously increased until the reference quieting is produced. The number of microvolts required to produce the reference shall be recorded to determine compliance with 3.9.21. The loading effect of the generators shall be considered when arriving at the number of microvolts required to produce the reference.

4.9.22 Oscillator radiation. The receiver shall be operated under standard test conditions except that no input signal shall be applied. A field strength meter, or equivalent, shall be connected to the antenna terminals of the receiver. The receiver shall be tuned to 30.00 MHz and scanned with the field strength meter. The frequencies and amplitudes of detectable outputs greater than 50 microvolts shall be recorded to determine compliance with 3.9.22. This procedure shall be repeated for dial settings of 46.50, 53.20, and 75.60 MHz.

4.10 Air-seal test. The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. The gage used for measurement of the vacuum shall be of such accuracy that any decrease in vacuum can be readily determined. The radio set shall comply with 3.6.

4.11 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252, MIL-STD-454, MIL-P-11268 and Drawing and Data List SC-DL-414900 (see 3.4, 3.5, and 3.15).

4.12 Inspection by gages. The mechanical dimensions and electrical performance, required by the mechanical and electrical test fixture drawings, shall be measured to determine conformance with the physical and functional requirements of 3.7. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect. The following dimensions or performance shall be gaged or measured:

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a. Mechanical (dimensions). The gage points covered by the Drawing and Gage List specified in (1) below and the additional gage points specified in (2) below:

(1) SC-GL-57991

(2)(a) The location and relationship to each other of pin and mounting screw holes on the printed circuit boards for all plug-in modules.

(b) The location in the case of the 6 mounting screw holes for the top cover.

(c) The location in the case of the 6 mounting screw holes for the bottom cover.

b. Electrical performance (modules). Prior to assembly into the receiver, modules shall be tested to determine compliance with the performance requirements specified in Drawing and Gage List SC-GL-323124 applicable to Receiver, Radio R-442( )/VRC.

4.13 500 hour operational test. The receiver shall be operated continuously for 500 hours at room temperature and with an input voltage of 30.0 vdc. At each 100 hour interval, the tests of 4.13(e), 4.13(f), and 4.13(g) shall be performed to determine compliance with 3.11. At the conclusion of the 500 hour test, without replacement of parts or realignment other than adjustment of front panel controls, the following tests shall be performed to determine compliance with 3.11.

a. Loudspeaker audio output (see 4.9.1.2).

b. Headphone audio output (see 4.9.1.3).

c. Monitor audio output (see 4.9.1.4).

d. Distortion, narrow band (see 4.9.2.1)

e. Sensitivity (see 4.9.5).

f. Squelch sensitivity, tone (see 4.9.6.1).

g. Squelch sensitivity, noise (see 4.9.6.2)

h. Catching range (see 4.9.7).

i. Input voltage (see 4.9.11).

4.14 Operational inspection. The receiver shall be operated at 30.05 and 60.0 MHz with standard test conditions, except that both the old and new squelch on positions shall be used with rf inputs of 0.7 and 0.5 microvolts, respectively. At each test frequency (30.05 and 60.0 MHz), a listening test shall be performed. Each radio set shall comply with 3.12.

4.15 Electromagnetic interference (EMI). The equipment shall comply with 3.13 using the test methods and procedures of Notice 3, MIL-STD-462.

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5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired levels(s) of protection shall be as specified by the acquisition activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Receiver, Radio R-442( )/VRC is intended for use in forward area combat vehicles. It can be used by itself or as an auxiliary receiver in Radio Sets AN/VRC-12( ), AN/VRC-44( ), AN/VRC-47( ), and AN/VRC-48( ).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and any amendments thereto.

b. Type required.

c. Issue of DODISS to be cited in the solicitation, and if required the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. When first article is required (see 3.1).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Quantity of running spare parts required (see 3.14).

h. Notice 4, MIL-STD-461A is to be used for requirements in 3.13.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the Contracting Officer on the adequacy of corrective action (see 4.5.3.2).

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6.5 Location of operational inspection and air-seal test. It is desirable that the operational inspection (4.14) and the air-seal test (4.10) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be samples and inspected immediately prior to packaging.

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: R-442W/VRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this document and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.8 Subject term (keyword) listing.

Frequency-modulated
Production burn-in plan
Auxiliary receiver

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX

PRODUCTION BURN-IN (DEBUGGING) PLAN
FOR
RECEIVER, RADIO R-442( )/VRC

10.0 SCOPE.

10.1 Scope. This plan defines the requirements, methods and procedures applicable to the burn-in (debugging) of Receiver, Radio R-442( )/VRC. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20.0 APPLICABLE DOCUMENTS.

The following specification forms a part of this appendix to the extent referenced herein.

Military

MIL-STD-781 Reliability Tests: Exponential Distribution

30.0 REQUIREMENTS.

Each Receiver, R-442( )/VRC on contract and first article shall be subjected to the burn-in process prior to Group A quality conformance inspection. When the performance measurements called for herein indicate a failure, the units will be repaired as required to eliminate the failure and returned for continuation of the burn-in process. Units submitted for acceptance shall have completed at least 42 hours of burn-in processing. Modules used for repair shall have completed at least 28 hours of failure free operation at the end of the 42 hour unit burn-in.

30.1 Test procedures.

30.1.1 Performance parameters to be measured. Audio Output Power of the units in Burn-in processing shall be continuously monitored to detect any cessation of performance or major degradation below specified limits. When a failure is detected by the monitoring system, the unit shall be repaired and retested to the extent deemed necessary by the Government Inspector. In addition, the following tests shall be performed on each unit prior to and following burn-in processing:

	Requirement	Test
Audio Power Output	3.9.1	4.9.1
Muting	3.9.1.1	4.9.1.1
Loudspeaker	3.9.1.2	4.9.1.2
Headphone	3.9.1.3	4.9.1.3
Monitor	3.9.1.4	4.9.1.4
Distortion, Narrow band	3.9.2.1	4.9.2.1
Sensitivity	3.9.5	4.9.5

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Squelch	3.9.6	4.9.6
Tone	3.9.6.1	4.9.6.1
Noise	3.9.6.2	4.9.6.2

30.2 Environmental and operational conditions.

30.2.1 Temperature test conditions. All cycling and testing shall be done at room ambient temperature, except that the temperature shall not be lower than +70° F.

30.2.2 Vibration cycling. Vibration cycling shall be in accordance with test level "B" of MIL-STD-781. (2.2G ± 10% at any non-resonant frequency between 20 and 60 Hz measured at the equipment mounting points and applied for 10 minutes out of each hour of "ON" time).

30.2.3 Input voltage cycling. The input voltage shall be varied in accordance with the following:

Nominal Voltage (25.5 VDC)	- 50% of the time
High Voltage (30.0 VDC)	- 25% of the time
Low Voltage (22.0 VDC)	- 25% of the time

This cycling procedure shall be repeated continuously throughout the burn-inprocess.

30.2.4 ON-OFF cycling. All units in Burn-in processing shall be subjected to ON-OFF cycling at least once during every 4 hour period and shall remain in the OFF condition for at least 5 minutes but not more than 15 minutes. Time spent in the OFF condition shall not be used in calculating the total Burn-in processing time specified in 30.0.

30.3 Applicable conditions.

30.3.1 Test scheduling. Testing shall be done on a 24 hour, 7 days per week basis.

30.3.2 Duty cycle. The radio set shall be operated continuously.

30.3.3 Maintenance. No adjustment or repair shall be made on the radio set except that necessary to enable a failed radio set to perform as specified. Operator controls shall be adjusted as required.

30.4 Burn-in test plan. A detailed Burn-in Test Plan shall be submitted for approval as specified in the contract. This test plan shall include, but not be limited to the following:

Detailed test procedures.
List of commercial test equipment used.
Schematics and complete functional descriptions of special test equipment.
Block diagrams of equipment interconnections.

30.5 Documentation failure reporting and analysis. Records shall be

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maintained on all units and modules to demonstrate to the Government Inspector that units presented for acceptance have complied with this document. In addition, a monthly report shall be submitted to the Government activity indicated in the bid request and contract (if required), categorizing all failures by types, numbers and cycle, and furnishing an analysis as to the reasons for the failures and corrective actions taken or recommended. This report shall be submitted no later than 30 days after the month covered and shall be furnished in five copies.

40.0 PERFORMANCE

40.1 Test procedures. Each unit under test shall be subjected to the testing specified in 30.1.1. Testing shall be conducted in accordance with the applicable referenced test paragraphs and each unit shall meet the limits specified in the applicable referenced requirement paragraph. Testing shall be done under the Standard Test Conditions of 4.8 (Receive), unless otherwise specified herein. Repeat all tests at 22 volts input voltage.

40.1.1 Special test equipment required.

Monitoring Console	-	Capable of detecting failure in Audio Output Power
Vibration Tables	-	Capable of operating in accordance with MIL-STD-781

40.2 Environmental and operational conditions. The Receiver, Radio R-442( )/VRC (inclosed in its case) shall be subjected to the following conditions:

40.2.1 Vibration Cycling. Vibration cycling shall be in accordance with 30.2.2 and MIL-STD-781 (Test Level B). The vibration table shall be located within the heat chamber.

40.2.2 Input voltage cycling. A 5 ampere regulated power supply shall provide the input voltages of 22-30 volts dc as specified in 30.2.3

40.2.3 ON-OFF cycling. ON-OFF cycling, as specified in 30.2.4, shall be accomplished by removing input power to all units.

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Custodian:
Army - CR

Preparing activity:
Army-CR
(Project 5820-A883)

User Activities:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-55099E(CR)	4 October 1990

3. DOCUMENT TITLE RECEIVER, RADIO R-442()/VRC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Commander, U.S. Army Communications-Electronics Command	(908)532-5851	992-5851
c. ADDRESS (Include Zip Code) ATTN: AMSEL-ED-TO(Stdzn) Fort Monmouth, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

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